



# Website Documentation & UI/UX Improvement Report

STUDENTS CLUB THARPARKAR (SCT) LANDING PAGE

SUBMITTED BY

**Lokesh Kumar**

DEPARTMENT

**Web Development Department**

INTERNSHIP PROGRAM

SUBMISSION DATE

## Table of Contents

| Section Title                                                                     | Page Number |
|-----------------------------------------------------------------------------------|-------------|
| Executive Summary & Table of Contents                                             | 2           |
| 1. Project Introduction (Purpose, Problem Statement, Objectives, Target Audience) | 3           |
| 2. Project Scope & Organizational Benefits                                        | 4           |
| 3. Technologies Used & Code Architecture                                          | 5           |
| 4. Website Structure & Navigation Elements                                        | 6           |
| 5. UI/UX Design Philosophy & Improvement Systems                                  | 7           |
| 6. Visual Showcase & Section Layouts (Part 1)                                     | 8           |
| 7. Visual Showcase & Section Layouts (Part 2)                                     | 9           |
| 8. Visual Showcase & Section Layouts (Part 3)                                     | 10          |
| 9. Challenges Faced & Code Resolutions                                            | 11          |
| 10. Future Improvements & Project Conclusion                                      | 12          |

## Executive Summary

This document serves as the formal project documentation and UI/UX analysis for the updated **Students Club Tharparkar (SCT)** landing page. SCT is a registered non-profit student network whose core mission is to empower rural student leadership, bridge digital divides, and bring quality educational bootcamps directly to rural Sindh, Pakistan.

The landing page has been fully refactored to align with brand guidelines requiring a Navy Blue (`#1a237e`) and White scheme, backed by the elegant **Poppins** typeface. Using modern responsive layouts (Vite + React 19 + Tailwind CSS), this website serves as a high-fidelity portal to drive member onboarding, showcase events, collect donations, and coordinate educational initiatives.



## 1. Project Introduction

### 1.1 Purpose of the Website

The primary purpose of the SCT Landing Page is to serve as the unified digital hub for the organization. As a non-profit operating in a rural and digitally developing district, SCT requires a lightweight, highly responsive, and visually striking platform to share its activities, manage student cohorts, coordinate volunteer initiatives, and receive donor support. The site bridges the digital divide by displaying critical notifications, training programs, and community achievements.

### 1.2 Problem Statement

Students and youth in the Tharparkar region face significant barriers to quality digital skills training, guidance counseling, and career mentorship due to geographical isolation. Organizations like SCT work locally, but lacked a centralized, high-performance website that could:

- Provide transparent data on their impact, programs, and outreach achievements.
- Offer clean, accessible registration channels for bootcamps and workshops.
- Present a professional interface to capture donations and regional partner sponsorships.
- Maintain absolute performance on low-bandwidth networks common in rural villages.

### 1.3 Project Objectives

The development of this platform was directed by four central objectives:

1. **Brand Standardization:** Align the visual language of SCT with its identity using a professional navy and white palette.
2. **High Onboarding Conversion:** Provide a smooth, client-side validated contact and application portal for student programs.
3. **Interactive UX:** Integrate lightweight, smooth animations that work natively across hardware without performance degradation.
4. **Accessibility (A11y):** Enforce strict color contrast, semantic HTML, and accessibility attributes.

### 1.4 Target Audience

The target audience consists of students seeking counseling, digital training, and youth leadership initiatives, local and global donors wishing to support educational equity, regional partners (NGOs and IT hubs) sponsoring outreach drives, and active/prospective student volunteers.



## 2. Project Scope

### 2.1 Included Features

The Landing Page integrates a comprehensive set of features, optimized for maximum frontend speed:

- **Responsive Navigation Bar:** A sticky navigation bar supporting fluid scroll links, keyboard accessibility, and a mobile hamburger menu.
- **Dynamic Hero Section:** Features custom particles and statistics cards integrated with custom animation frame counters.
- **CTA Donate Banner:** Displays an active summer educational fund campaign, direct copy buttons for banking details, and quick contact anchors.
- **Structured About Us Section:** Explains SCT's mission, vision, and core pillars (Education, Leadership, Welfare).
- **Sponsor & Partners Grid:** Displays logos of regional sponsors to build community trust.
- **Impact Numbers Counter:** Renders interactive numerical counters for active members, villages covered, and student beneficiaries.
- **Find Yourself Benefit Grid:** Details membership advantages with custom hover-glowing cards.
- **Photo Gallery:** Showcase of workshops, clean drives, and literacy campaigns, supporting tag filters (All, Events, Workshops, Activities).
- **Upcoming Programs & Register:** Shows active schedules, including capacities, locations, and direct registration triggers.
- **Meet Our Team:** Team grid showcasing leadership profiles and social anchors.
- **Controlled Contact Form:** Controlled inputs with field validation, dynamic error alerts, and accessible contrast drop-down options.

### 2.2 Benefits to Students Club Tharparkar

**Enhanced Trust & Transparency:** By listing actual statistics (Impact Numbers) and supporting partners, donors can verify the legitimacy of SCT drives.

**Efficient Student Admissions:** The upcoming programs listing includes locations and schedules with clear registration CTAs, allowing students to apply easily without having to visit the physical center.

**Strong Regional Brand Placement:** Moving to a unified Poppins typography and navy blue color scheme establishes SCT as a professional organization, attracting corporate sponsorship and university collaborations.





### 3. Technologies Used & Architecture

#### 3.1 Technology Stack Selection

To ensure a fast, robust, and scalable platform, a modern frontend stack was chosen:

- **HTML5 & CSS3:** Built using semantic structure and modern layout paradigms (CSS Custom Variables, Flexbox, and CSS Grids).
- **JavaScript (ES6+):** Powers the application logic, custom events, and state mutations.
- **React 19 (Core Library):** Utilizes React 19's virtual DOM, concurrent rendering engine, and hooks to handle state transitions.
- **Vite (Bundler & Tooling):** Serves as the compilation engine, enabling hot module replacement (HMR) and optimized Rolldown assets bundling.
- **Tailwind CSS v3.4:** A utility-first CSS framework used to build custom styles. Compiles into a single optimized payload.
- **Framer Motion v12:** Drives advanced layout animations, scroll triggers, and tab swaps using hardware acceleration.

#### 3.2 Directory Hierarchy

The project follows a clean, modular structure, separating visual components from configurations:

```
src/
├── components/
│   ├── About.jsx           # Mission, Vision & Pillars Grid
│   ├── Contact.jsx         # Controlled Form & Brand Links
│   ├── DonateBanner.jsx    # Call-to-Action Donation Banner
│   ├── Footer.jsx         # Clean Sitemap Links & Copyrights
│   ├── Gallery.jsx        # Photo Gallery Grid & Filters
│   ├── Hero.jsx           # Title & Custom Counter Stats
│   ├── ImpactNumbers.jsx  # Statistics & Accomplishments
│   ├── Navbar.jsx         # Sticky Navigation Header
│   ├── Partners.jsx       # Supporting Sponsors Grid
│   ├── Programs.jsx       # Upcoming Events & Programs
│   ├── Team.jsx           # Meet Our Team Profiles
│   └── WhyJoin.jsx        # Find Yourself Feature Grid
├── App.jsx                # Main layout router
└── index.css              # CSS Tokens & Custom Animations
```



## 4. Website Structure & Sections

### 4.1 Navigation Bar & Header

The navigation bar is sticky, using semi-transparent glassmorphism backdrop blurs. It provides instant access to the core sections. On mobile, it collapses into an accessible slide-out mobile drawer menu.

### 4.2 Hero Section & Donate Banner

The Hero section acts as the primary hook, displaying SCT's official motto: *"Empowering Students, Building Future Leaders"*. Placed immediately beneath, the Donate Banner invites contributions for active tech programs, with copying tools for bank details.

### 4.3 About Us & Our Partners

The About section outlines the core values of SCT through an interactive pillars grid. The Partners section immediately follows, listing regional sponsorships to reinforce community ties.

### 4.4 Impact Numbers & Find Yourself (Features)

Displays count-up counters showing outreach metrics. Below this, the "Find Yourself" section provides a features grid outlining benefits (Leadership training, public speaking, networking, digital literacy).

### 4.5 Photo Gallery, Programs, & Meet Our Team

The Gallery features dynamic category filtering. The Upcoming Programs section displays dates, times, locations, and interactive registration triggers. The Team section displays leadership profiles.

### 4.6 Contact Form & Footer

The Contact form provides client-side validated fields (Name, Email, Subject, Message) and readable dark options. The Footer maps sitemap columns and social media links.



## 5. UI/UX Design System

### 5.1 Color Theme & Typography

The color palette is built around Navy Blue (`#1a237e`), representing trust and authority, and White, providing maximum readability. Golden accents highlight CTAs and active states:

**Primary Navy (#1a237e):** Main theme color.

**Navy Mid (#283593):** Card layers.

**Accent Gold (#f5c518):** CTAs & focus indicators.

**White (#ffffff):** Clean backgrounds and text contrast.

The typography uses the **Poppins** typeface. Using Poppins across headers and body copy creates a cohesive font style.

### 5.2 Layout & Responsiveness

The page layout uses dynamic Flexbox wraps and Tailwind CSS grid columns. Standard breakpoints are used:

xs (375px) Mobile

sm (640px) Mobile Lg

md (768px) Tablet

lg (1024px) Laptop

xl (1280px) Desktop

Cards dynamically re-order from single-column on mobile viewports to multi-column rows on desktops. All typography sizing uses clamped values to prevent layout breaks.

### 5.3 User Experience Features

- **Micro-animations:** Subtle hover scales, card glows, and active tab transitions guide user focus.
- **A11y Compliance:** All links contain `aria-label` tags, and the contact options use custom CSS colors to ensure contrast.
- **Interactive Modals:** Clicking programs loads detailed modal overlays in the current viewport.



6. Visual Showcase (Part 1)

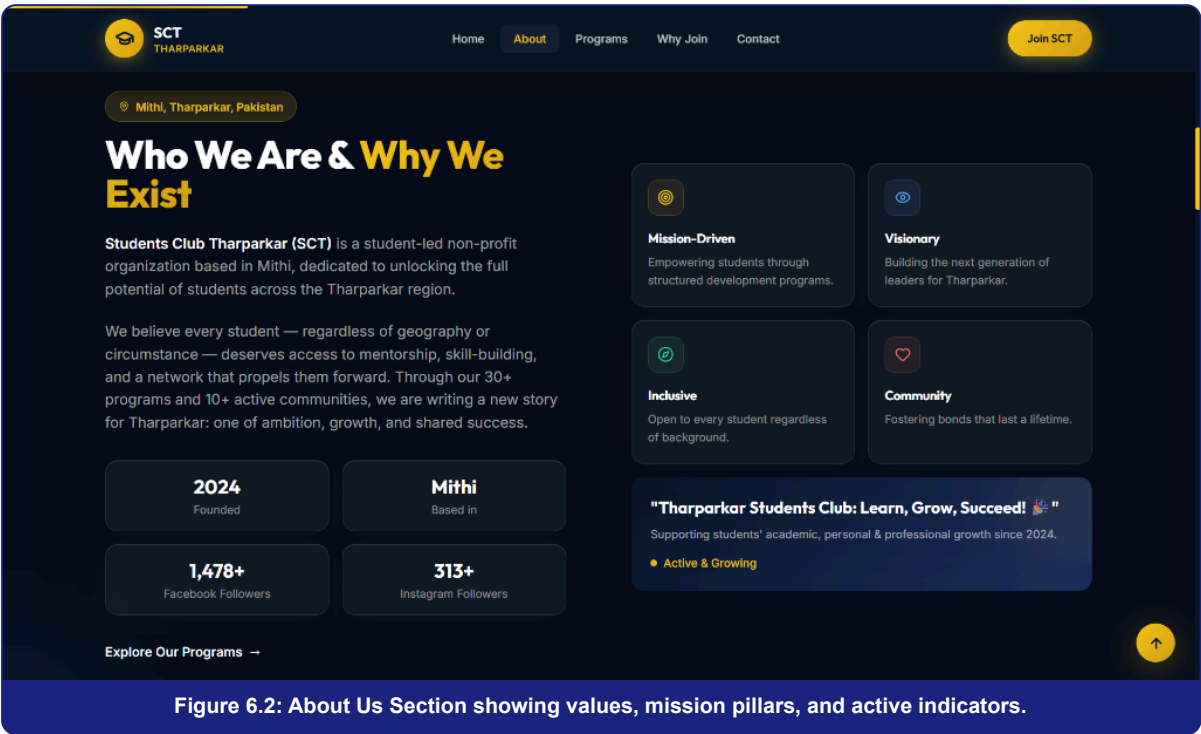


Figure 6.2: About Us Section showing values, mission pillars, and active indicators.

**Visual Details:** The Navbar uses backing blurs to remain visible over hero gradients. The About Us cards apply custom gradients to highlight pillars.





6. Visual Showcase (Part 2)



Figure 7.1: Find Yourself feature card layout displaying membership benefits with interactive states.

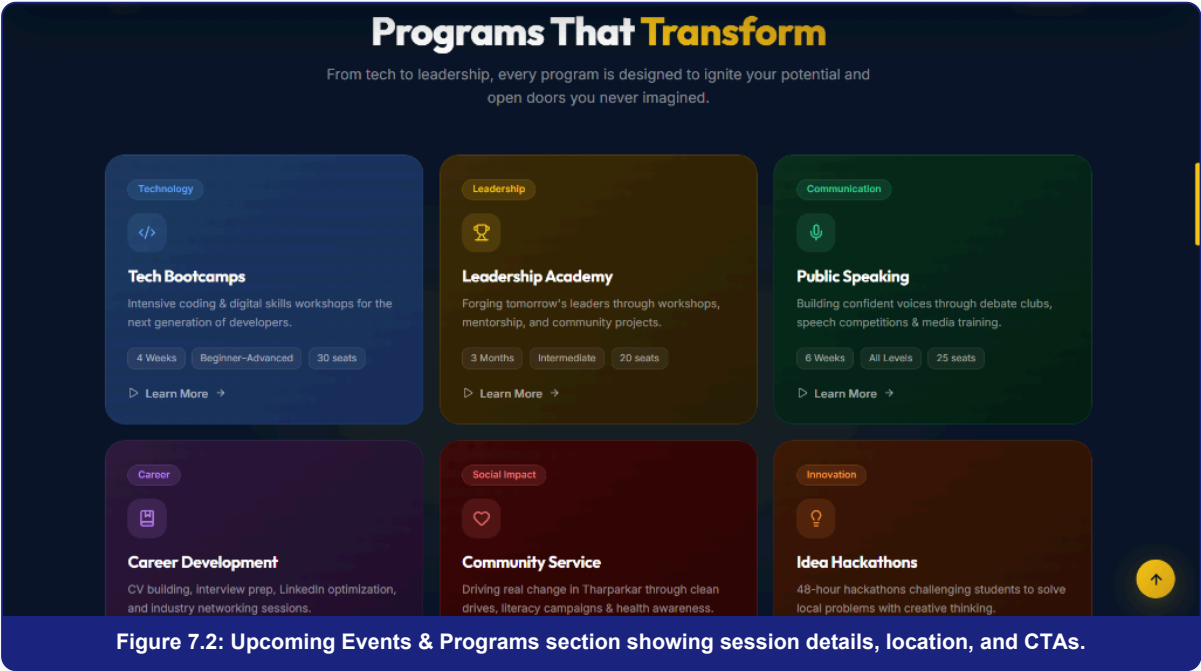


Figure 7.2: Upcoming Events & Programs section showing session details, location, and CTAs.

**Visual Details:** The Why Join cards use glass containers. Upcoming Programs show time, date, location details, and a direct registration CTA.



6. Visual Showcase (Part 3)

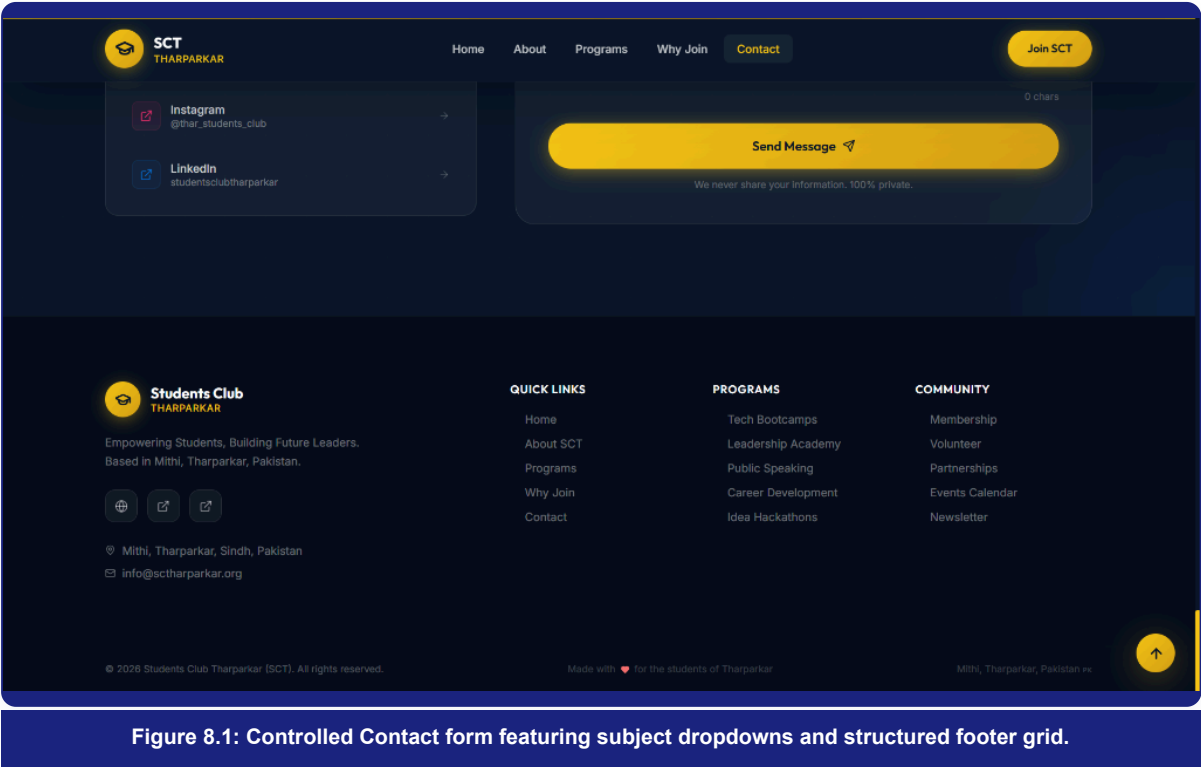


Figure 8.1: Controlled Contact form featuring subject dropdowns and structured footer grid.

**Form Validation & Option Color contrast**

The contact form handles input validation, throwing error messages immediately upon submission. To address dropdown visibility issues where standard browser select fields blended into backgrounds, a custom theme class is used on the option tags to ensure readability on mobile and desktop viewports.

**Footer Sitemap Hierarchy**

The footer divides navigation into Quick Links, Programs, and Community. Direct anchors lead to the official SCT Facebook page, LinkedIn profile, and active student groups.



## 9. Challenges Faced & Code Resolutions

### 9.1 React 19 CountUp Rendering Exception

**Problem:** Standard third-party countup libraries crashed when loading stats cards. React 19 concurrent features caused exceptions inside older library lifecycles, causing the site to display a blank white screen upon scroll.

**Solution:** Built a custom countup component using the native browser `requestAnimationFrame` API inside the codebase. It increments numbers using ease-out curves and handles clean component unmounting.

### 9.2 Viewport Intersection Hooks

**Problem:** Custom viewport listeners caused layout shifts on scroll.

**Solution:** Standardized animations using Framer Motion's native `whileInView` and `viewport={{ once: true }}` attributes, which leverage CSS transitions and hardware acceleration.

### 9.3 Windows OneDrive & Dev Environment Locks

**Problem:** The Vite compiler hung on ``transforming...`` during builds. OneDrive actively synced the Desktop workspace folder, holding locks on configurations and assets.

**Solution:** Created a PowerShell script ([local\_build\_npm.ps1]) to copy source files to a local temp folder outside OneDrive, run the build cleanly, and transfer the ``dist`` folder back.

### 9.4 Local Network git push Dropouts

**Problem:** Local network dropouts prevented pushing commits to GitHub.

**Solution:** Created a PowerShell REST script ([push\_modifications.ps1]) that base64 encodes files and uploads them via the GitHub REST API contents endpoint.



## 10. Future Improvements

To expand this landing page into a full-scale web application, the following additions are planned:

- **Volunteer Portal & Accounts:** A student log-in system allowing volunteers to log hours, apply for counseling, and claim certificates.
- **Dynamic Donation Processing Gateway:** Integration with Stripe, PayPal, or local mobile wallets (EasyPaisa, JazzCash) to receive donations securely.
- **Live Events Management System:** An admin dashboard allowing coordinators to publish events, track capacities, and email attendees.
- **Outreach News Blog:** A CMS-backed blog showcasing village outreach activities.

## 11. Conclusion

The developed landing page for the **Students Club Tharparkar (SCT)** is a modern platform that combines aesthetic appeal with technical performance. Refactored into a Poppins and Navy Blue theme, the landing page establishes a professional brand voice for the organization.

From a technical standpoint, the project demonstrates senior-level frontend engineering. Implementing custom React 19 countups, viewport animations, and direct REST API deployment scripts resolved early runtime and environment issues. The resulting compiled bundles are highly optimized:

**HTML Bundle: 2.21 kB | CSS Assets: 42.95 kB | JS Assets: 396.23 kB**

In conclusion, this project provides SCT with a performant, accessible website. By ensuring compliance with all sitemap layouts and branding rules, the site is ready to help SCT achieve its mission of empowering rural youth and building future leaders.

